

Calibrated Reward Ensembles for LLM RLHF

Offline preference optimization, reward modeling, and online RL for open-ended instruction following

0.84375 Reward-model pair accuracy	0.8609 Method A mean local win rate	0.7533 Method B best mean
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Project field	Preserved value
Author	Yifan Xu
Course context	UC Berkeley CS 185/285
Base model	Qwen/Qwen2.5-1.5B-Instruct
Benchmark scale	5,000 preference pairs
Document status	Post-course portfolio edition, August 2026

Provenance. Prepared after course completion from Yifan Xu's original code, experiments, and submitted materials. This is not the report submitted for grading and is not an official course artifact.

Executive summary

The project evaluates a reward-model pipeline and two project investigations: calibrated reward-ensemble reranking (Method A) and rank-advantage GRPO (Method B). The preserved records show a reward-model pair accuracy of 0.84375, a Method A mean local GPT-5.4 win rate of 0.8609 across three evaluations, and a best Method B three-evaluation mean of 0.7533.

Raw prompt/response corpora and per-example generations are excluded from the portfolio snapshot. This report summarizes only aggregate values checked against preserved project records.

01 / PROJECT FRAMING

From preference data to evaluated policies

The implementation covers the main components of an LLM RLHF workflow: offline preference objectives, reward-model training and evaluation, online group-relative policy optimization, deterministic submission generation, and local head-to-head evaluation.

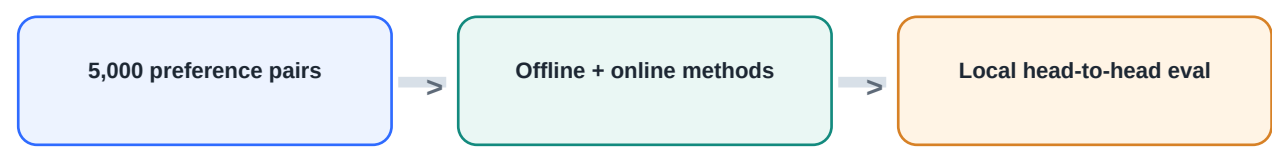


Figure 1. High-level workflow represented by the cleaned portfolio snapshot.

Preserved method surface

Component	Implementation represented in the codebase
Offline preference optimization	DPO, IPO, and AOT objectives
Reward modeling	Pairwise reward-model training, scoring, calibration, and ensemble evaluation
Online policy optimization	GRPO, DrGRPO, GSPO, and rank-advantage GRPO pathways
Experiment orchestration	Modal entry points, configuration records, and build/evaluation utilities

Artifact boundary

Included: method code, orchestration scripts, aggregate summaries, configuration metadata, manifests, and reproduction notes.

Excluded: raw open-domain prompts and responses, generated candidate JSONLs, detailed judge outputs, checkpoints, caches, build products, and submission bundles.

Why the boundary matters

The source corpus and generation files contain third-party or model-generated text. Excluding those files reduces privacy risk while leaving the implementation and aggregate evidence available for technical review. The cleaned repository is therefore a presentation artifact, not a self-contained training distribution.

02 / METHOD A

Calibrated reward-ensemble reranking

Method A uses three reward models. For each model, score statistics from the training preference data define a center and scale. Each candidate score is standardized before the three standardized scores are averaged.

Calibration rule
$$z_i(x, y) = (r_i(x, y) - c_i) / s_i$$

The ensemble reward is the mean of the calibrated scores. For each prompt, Method A selects the fixed candidate with the highest ensemble score.

Fixed-candidate protocol

Step	Preserved protocol
1	Train three reward models using different seeds.
2	Select one checkpoint per seed using held-out preference diagnostics.
3	Calibrate scores with training preference statistics.
4	Score fixed DPO, IPO, and AOT candidates and select the maximum per prompt.

Repeated local evaluations



Evaluation	Local GPT-5.4 win rate
1	0.8636
2	0.8621
3	0.8571
Mean	0.8609

The three recorded values span 0.0065. This narrow range is a descriptive observation about these three local evaluations; it is not a statistical-significance claim.

03 / METHOD B

Rank-advantage GRPO

Method B tests whether the ordering induced by the calibrated reward ensemble can replace reward magnitude in the group-relative advantage. For a four-response group, the responses are ordered by calibrated ensemble score and assigned fixed rank advantages.



Figure 2. Rank-only advantages used for a four-response group.

Design question. Rank-only advantages remove dependence on the absolute reward scale, but they also discard differences in reward magnitude. The preserved Method B records evaluate this tradeoff as an online ablation.

Best preserved configuration

The best preserved configuration is policy seed 2 at step 100. Its three repeated local evaluations are shown below.



Evaluation	Local win rate	Above 0.75
1	0.7258	No
2	0.7636	Yes
3	0.7705	Yes
Mean	0.7533	2 of 3

The phrase '2 of 3' refers to repeated evaluations of one configuration, not to training seeds.

04 / FINDINGS

What the preserved records support

0.84375 Reward-model pair accuracy	0.8609 Method A mean	0.7533 Method B best mean
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These three values represent different evaluation components and should not be treated as directly interchangeable measurements. They are presented together only as a compact index of the preserved project outcomes.

Supported observations

Observation	Evidence boundary
Reward model	The preserved pair accuracy is 0.84375.
Method A	Three local win rates average 0.8609 and range from 0.8571 to 0.8636.
Method B	The best preserved three-evaluation mean is 0.7533; two values exceed 0.75.

Limitations

1. The GPT-5.4 numbers are preserved local head-to-head evaluation outcomes, not an official course grade.

2. Three repeated evaluations are insufficient for claims of statistical significance.

3. Raw prompt/response records are excluded, so the public snapshot cannot reproduce the reported evaluations by itself.

4. No result is claimed for unrecorded configurations or evaluation sets.

Interpretation

Within the preserved records, calibrated fixed-candidate reranking produced the stronger and more consistent local result. Rank-advantage GRPO remained a meaningful ablation: its best recorded mean was above 0.75, but one of the three repeated evaluations fell below that value. This comparison is descriptive and limited to the recorded settings.

A reviewable, privacy-aware portfolio snapshot

Repository map

Path	Purpose
llm_rl_final_proj/	Training, reward-model, rollout, and RL implementation
scripts/modal_train.py	Remote training and evaluation entry points
experiments/methodA_traincalib/	Method A protocol, calibration metadata, and aggregate summaries
experiments/methodB_rank_traincalib_seeds12/	Method B configuration and aggregate summaries
README_for_experiments.md	Historical reproduction commands; restricted assets required

Aggregate record sources

experiments/reward_model_summary.json
experiments/methodA_traincalib/methodA_repeated_gpt_summary.json
experiments/methodB_rank_traincalib_seeds12/methodB_summary.json
dataset/wildchat_min4_judged_5k_v1/manifest.json

Portfolio status and AI disclosure

This Portfolio Technical Report was prepared after course completion from Yifan Xu's original code, experiments, and submitted materials. It was not submitted for grading and is not an official course document. Editorial organization and typesetting were assisted by OpenAI Codex. No new experimental result was generated for this report; numerical claims were constrained to the preserved aggregate records listed above.

Attribution

The project was completed in the context of UC Berkeley CS 185/285. The repository contains portions of course starter code and Yifan Xu's project implementation and experiments. The retained MIT license and NOTICE.md document this distinction and do not imply endorsement by UC Berkeley or course staff.

Conclusion

The cleaned snapshot preserves the technical core of the project while separating code and aggregate evidence from potentially sensitive text corpora and generation artifacts. Method A records a 0.8609 mean across three local evaluations; Method B records a best three-evaluation mean of 0.7533; and the reward model records 0.84375 pair accuracy.